Measurement of the Drift of Charge Carriers in a Semiconductor Using the Transient-Current Technique

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Abstract

This study investigates the drift dynamics of charge carriers in a 300 μm thick silicon semiconductor using the Transient-Current Technique (TCT). By irradiating the sensor with a 600 nm red laser on both sides, electron and hole dominated transient signals were isolated and recorded as a function of applied reverse bias voltage and laser intensity. The total collected charge was analytically computed by integrating the averaged waveforms and compared to direct gated-charge measurements, confirming full signal efficiency in the overdepletion regime and a linear dependence on laser intensity. Finally, the charge carrier transit times were extracted from the signal durations to estimate the effective mobilities. The analysis yielded $\mu_h = (527 \pm 102)\ cm^2/(V \cdot s)$ for holes, consistent with theoretical expectations for high-purity silicon, and $\mu_e = (1203 \pm 139)\ cm^2/(V \cdot s)$ for electrons. The relatively high uncertainties and slight deviations from ideal values are attributed to the limited data points, localized doping variations, and potential temperature fluctuations during the measurements.

Introduction

The primary aim of this experiment is to investigate the drift dynamics of charge carriers, specifically electrons and holes, within a silicon semiconductor detector utilizing the Transient-Current Technique [1].

This report is structured to provide a comprehensive overview of the experiment, going from the underlying physical principles to the final results. Section 1 introduces the theoretical background, detailing the physics of semiconductors, the equations governing the drift motion of charge carriers, and the physical mechanisms of signal formation. Section 2 describes the experimental setup, outlining the hardware components, the laser apparatus, and the data acquisition procedure utilized in the laboratory. In Section 3, the collected data is presented and systematically analyzed; this includes the study of signal waveforms under varying bias voltages and laser intensities, charge calculations, and the analytical estimation of the carrier mobilities. Finally, Section 4 summarizes the main findings of the experiment and draws the final conclusions.

1 Theoretical Background

1.1 Semiconductors in Particle Physics

Semiconductors, as the name suggests, have electrical properties that sit right between a conductor and an insulator. This behavior is fundamentally linked to their band gap, which goes from $\sim 0.1 - 0.2$ eV to $6 - 7$ eV. Silicon, which is the most widely used material in the fabrication of semiconductor detectors, has a band gap of approximately 1.107 eV and crystallizes in a diamond lattice structure where each atom is covalently bound to four nearest neighbors (Figure 1).

At absolute zero, all electrons are tightly bound in the valence band. However, at room temperature, thermal energy is enough to kick some of these electrons across the gap and into the conduction band. When an electron makes this jump, it leaves behind an empty spot in the lattice. We call this vacancy a "hole", and for all practical purposes, it behaves as a positively charged quasi-particle with its own effective mass. Intrinsic (pure) silicon features a relatively low charge carrier density of about $10^{10}\,\mathrm{cm}^{-3}$ (compared to the $10^{22} - 10^{23}\,\mathrm{cm}^{-3}$ of conductors), which makes it a poor conductor by itself. To get around this and tailor the material for our needs, we intentionally introduce impurities into the crystal lattice in a process called doping. If we dope the silicon with atoms that have five valence electrons, the extra electrons are free to move around, creating an *n-type* semiconductor (N-type doping). Instead, if we use atoms with only three valence electrons, the lattice is "missing" electrons, meaning we get an excess of holes, and this gives us a *p-type* semiconductor (P-type doping). This doping process leads to unequal number densities of charge carriers, defining majority and minority carriers within the respective materials.

Particle detection happens when you put a p-type and an n-type semiconduc-

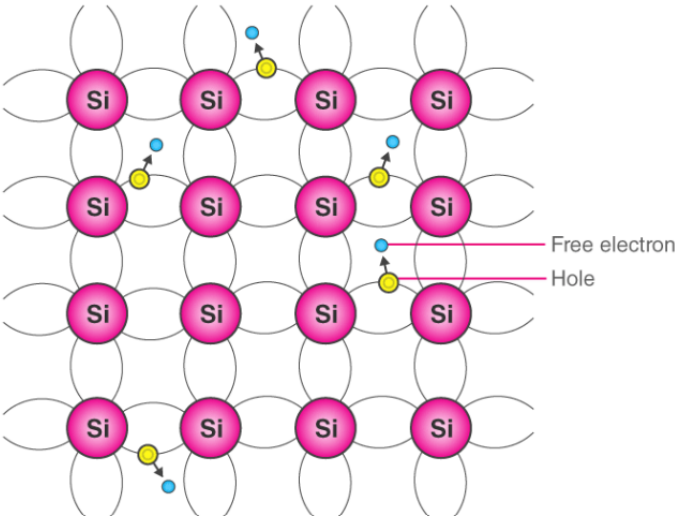


Figure 1: Schematic representation of the silicon crystal lattice and covalent bonds, with electrons and holes.

tor together to form a p-n junction. Because of the difference in concentration, electrons from the n-side naturally want to diffuse over to the p-side to fill the holes. As they do this, they leave behind fixed positive ions on the n-side and create fixed negative ions on the p-side. This charge build-up generates a built-in electric field that eventually stops further diffusion. The resulting area in the middle is left depleted of free charge carriers, and it is called the depletion region. When an ionizing particle (or a laser pulse) passes through this depletion region, it interacts with the silicon lattice and generates fresh electron-hole pairs along its track. The electric field immediately separates these charges before they can recombine, forcing them to drift toward the opposite electrodes and thereby inducing a detectable current signal. However, the natural depletion region is too narrow to be an efficient particle detector. To actually use this junction to detect particles, we need a much larger sensitive volume. We can achieve this by applying a reverse bias voltage across the sensor, as seen in Figure 2. This external voltage basically pulls the majority carriers further away from the junction interface, stretching the depletion region across the bulk of the material. The thickness d of this depleted zone is directly related to the applied bias voltage V and the effective doping concentration of the material, following the relation:

$$d(V) = \sqrt{\frac{2\epsilon(V + V_D)}{qN_{eff}}} \tag{1}$$

where ϵ is the dielectric constant of silicon, V_D is the built-in potential, q is the elementary charge, and N_{eff} is the effective doping concentration.

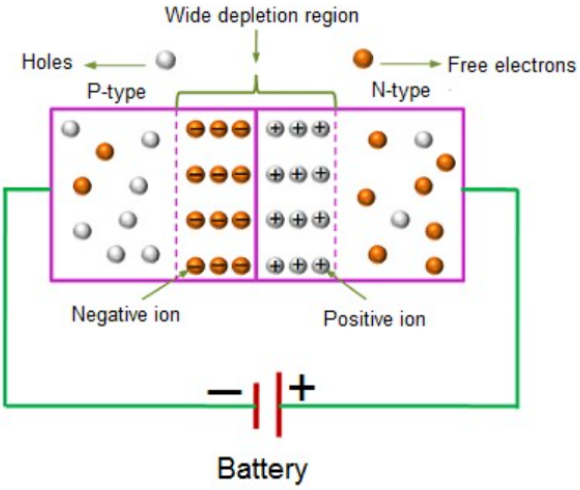


Figure 2: A p-n junction under reverse bias, highlighting the widening of the depletion region.

When a high enough voltage is applied, the sensor becomes fully depleted. Even with a strong reverse bias, thermal energy is always randomly generating a few electron-hole pairs in the depletion zone by making the lattice oscillate. The electric field moves these charges just like the signal charges, creating a continuous, annoying background current known as leakage current. This is why in many high-energy physics experiments, silicon detectors must be cooled down to keep this noise under control [2].

1.2 Drift Motion of Charge Carriers

Once the laser pulse creates the electron-hole pairs inside our depleted sensor, these charges need to travel to the electrodes so we can actually measure a signal. However, their journey isn't just a simple straight line. In reality, the charge carriers undergo two superimposed types of motion.

First, there's a random, chaotic thermal motion. Just by being at room temperature, the electrons and holes are constantly bouncing off the silicon atoms following a Maxwell-Boltzmann distribution. This thermal velocity is actually quite high, but since it's completely random, it averages out. The second type of motion is what we actually care about for our signal: the drift motion. Since we've applied a strong reverse bias voltage, there is a macroscopic electric field across the sensor. This field moves the electrons in one direction and the holes in the opposite one, imposing a directional drift over their chaotic thermal motion. In the silicon crystal, the charge carriers are accelerated by the electric field but are constantly scattered by lattice vibrations (phonons) and crystal defects.

We can describe this balancing act between acceleration and "friction" using the classic Drude model. If we look at the average drift velocity $\vec{v}_D$, the equation of motion is:

$$m^* \left(\frac{d\vec{v}_D}{dt} + \frac{\vec{v}_D}{\tau} \right) = q\vec{E} \tag{2}$$

where q is the charge, m^* is the effective mass of the carrier (which accounts for the quantum mechanical binding inside the lattice), $\vec{E}$ is our electric field, and τ is the collision time (or relaxation time). Basically, τ is the average time a carrier travels before bumping into the lattice and changing its momentum. In silicon, this happens incredibly fast, on the scale of picoseconds.

When the system reaches a steady state (which happens almost instantly), the acceleration term drops out, and the drift velocity becomes proportional to the electric field. This gives us the famous linear relation:

$$\vec{v}_D = \mu\vec{E} \quad \text{with} \quad \mu = \frac{q\tau}{m^*} \tag{3}$$

The proportionality constant μ is the mobility of the charge carrier. Since holes have a larger effective mass than electrons, their mobility is significantly lower, which is why we'll see them taking much longer to cross the sensor in our oscilloscope waveforms.

However, this linear relationship doesn't hold forever. It works fine at low electric fields, where the carriers mostly scatter by emitting low-energy acoustic phonons, keeping the system in thermal equilibrium. But in high electric fields, the carriers gain so much kinetic energy between collisions that they start exciting optical phonons (higher-energy lattice vibrations). This acts like a much stronger braking mechanism. The mobility μ isn't constant anymore; it drops as the field increases. Eventually, the drift velocity stops growing linearly and flattens out, approaching a maximum limit called the saturation velocity (v_{sat}).

To model this behavior over the entire range of voltages we use in the experiment, physicists rely on a well-known empirical formula:

$$v_D = \frac{\mu_0 E}{\left[1 + \left(\frac{\mu_0 E}{v_{sat}} \right)^{\beta} \right]^{1/\beta}} \tag{4}$$

where μ_0 is the low-field mobility and β is just a material-dependent parameter. Understanding this saturation effect is crucial for our experiment, because when we push the bias voltage to extreme values, we shouldn't expect the signal to just keep getting faster and faster infinitely [2].

1.3 Signal Formation and Duration

A common misconception in particle detection is that a signal is only recorded when a charge carrier physically reaches the readout electrode. However, the formation of the electrical signal is fundamentally governed by the Shockley-Ramo

theorem, which states that the instantaneous current induced on an electrode is caused by the movement of the charge carriers through the electric field, rather than their actual collection at the destination. Therefore, even if a charge gets trapped in the bulk before reaching the electrode, its preceding motion still induces a measurable signal.

In a simple parallel-plate detector filled with a uniform dielectric, the internal electric field is constant. However, a fully depleted semiconductor detector contains a spatially distributed space charge. Consequently, the internal electric field is not constant: it decreases linearly from the p-n junction side to the opposite contact. If the applied bias voltage V is larger than the depletion voltage V_{dep} (a condition known as overdepletion), the electric field $\vec{E}(x)$ across a sensor of thickness d can be described as:

$$\vec{E}(x) = -\left(\frac{V + V_{dep}}{d} - \frac{2V_{dep}}{d^2}x\right)\vec{e}_x = -(a - bx)\vec{e}_x \tag{5}$$

where we introduced the constants $a = (V + V_{dep})/d$ and $b = 2V_{dep}/d^2$ to simplify the mathematical notation,

When an electron-hole pair is generated at a specific position x_0, the two carriers begin to drift in opposite directions under the influence of this field. Assuming we operate in a regime where the mobility μ is roughly constant, the drift velocities become position-dependent due to the non-uniform electric field:

$$v_e(x) = -\mu_e E_x(x) = +\mu_e(a - bx) \tag{6}$$
$$v_h(x) = +\mu_h E_x(x) = -\mu_h(a - bx) \tag{7}$$

This indicates that the charge carriers experience an accelerated (or decelerated) motion as they transit through the sensor. By integrating these velocity equations, we can determine the time it takes for each carrier to reach their respective electrodes, which effectively corresponds to the duration of the induced signal.

For electrons reaching the electrode at $x = d$ and holes reaching the electrode at $x = 0$, the total transit times T^- and T^+ are:

$$T^- = \tau_e \ln\left(\frac{a - bx_0}{a - bd}\right) = \frac{d^2}{2\mu_e V_{dep}} \ln\left(\frac{V + V_{dep}}{V - V_{dep}}\right) \tag{8}$$
$$T^+ = \tau_h \ln\left(\frac{a - bx_0}{a - bd}\right) = \frac{d^2}{2\mu_h V_{dep}} \ln\left(\frac{V + V_{dep}}{V - V_{dep}}\right) \tag{9}$$

Here, $\tau_{e,h}$ represents the characteristic time for the charge carrier movements [2].

In the context of our TCT setup, a red laser deposits charge very close to the sensor surface. By irradiating the front or the back of the sensor, we effectively set the initial charge generation point x_0 to approximately 0 or d. This experimental technique allows us to isolate the signal of one specific charge

carrier traversing the entire bulk, enabling the measurement of its transit time and the subsequent estimation of its mobility with a fit, as we will see later.

2 Experimental Setup

The experimental apparatus utilized for this laboratory session is shown in Figure 3. The core of the system is a red laser with a wavelength of $\lambda = 600$ nm, which is used to irradiate the semiconductor sample[1]. The sample itself is a 300 μm thick silicon sensor. For this specific detector, the full depletion voltage is known to be (-14.83 ± 0.37) V. To ensure full charge collection and operate in an overdepleted regime, the sensor is reverse-biased using a high-voltage (HV) source. It is worth noting that while diode breakdown slowly begins around -200 V (leading to an increase in leakage current), a maximum bias voltage of up to -500 V can be applied to the system.

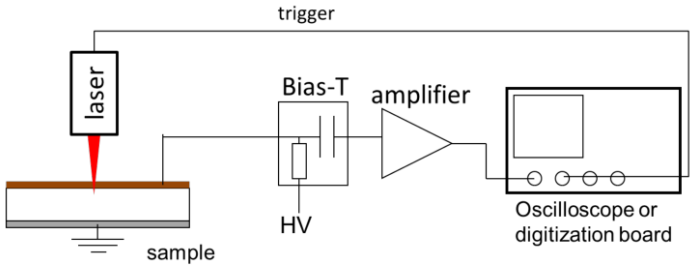


Figure 3: Schematic diagram of the Compact-TCT experimental setup, illustrating also an additional oscilloscope.

The high voltage is supplied to the sample through a Bias-T circuit. Once the charge carriers drift through the electric field, the induced current is routed through the Bias-T and fed into an amplifier. The amplified signal is then transmitted to a DRS board, which practically acts as a digitizing oscilloscope.

During the experiment, the sensor was physically flipped to repeat the measurements in both orientations. Because the laser generates charges strictly at the surface, one type of carrier is collected instantly while the other must cross the entire silicon bulk. This method effectively isolates the signals of electrons and holes, allowing us to study their transport properties independently. To properly analyze the temporal evolution of the signal and calculate the transit times, precise synchronization is required. For this reason, the digitization process is triggered directly by the synchronization output pulse of the laser driver, which is connected to the DRS board. Both the laser controller and the DRS board are interfaced with a PC via USB connections, allowing the user

[1] As anticipated before, red light has a very shallow penetration depth in silicon, so the electron-hole pairs are generated just below the illuminated surface.

to remotely adjust the laser intensity and frequency, as well as to record the digitized waveforms for offline data analysis.

3 Data analysis

3.1 Waveform Analysis

The digitized signals from the semiconductor sensor were processed by averaging 1000 measured waveforms for each experimental configuration. This was done to study the sensor's response under different bias voltages (keeping a fixed laser intensity of 2%) and different laser intensities (keeping a fixed bias voltage of -500 V). Furthermore, a Gaussian filter was applied in post-processing to reduce the high-frequency electronic noise and obtain cleaner, more legible waveforms.

The shape of the recorded waveforms differs significantly depending on the type of charge carrier crossing the bulk (holes or electrons), mainly due to their different effective masses and intrinsic mobilities.

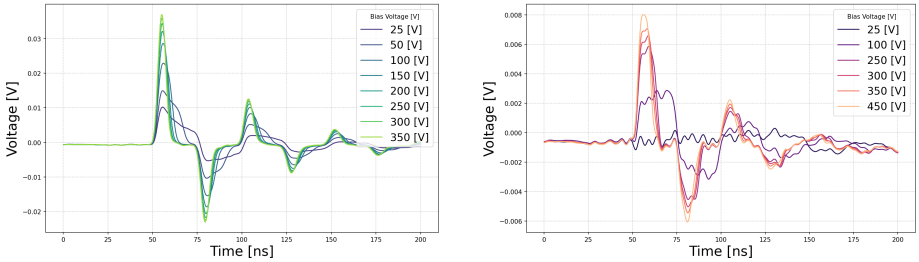


Figure 4: Mean waveforms recorded at varying bias voltages for both electron (on the left) and hole (on the right) transient signals.

As observed in Figure 4, increasing the magnitude of the reverse bias voltage has two main effects on the primary signal pulse (the first large peak): its amplitude increases, and its overall duration decreases. This behavior is perfectly aligned with the theoretical models discussed in the previous sections. In fact, a higher bias voltage generates a stronger internal electric field across the depleted sensor. This stronger field accelerates the charge carriers to higher drift velocities. Consequently, the charges cross the silicon bulk much faster, which explains the shorter temporal duration of the pulse. At the same time, according to the Shockley-Ramo theorem, the instantaneous induced current on the readout electrode is directly proportional to the velocity of the moving charges. Therefore, a higher drift velocity translates directly into a higher current amplitude. It is also worth noting that at the highest voltages (from 250 V and above), the amplitude increase becomes less pronounced; this happens because the carriers are approaching their saturation drift velocity.

A structural comparison reveals clear differences between the electron (left panel) and hole (right panel) waveforms. Electron signals exhibit significantly

higher amplitudes and shorter durations due to their lower effective mass and correspondingly higher mobility. Furthermore, their current peaks immediately at the beginning of the transient: electrons are generated near the readout electrode where the electric field is strongest, and they decelerate as they drift towards the lower-field backside. Conversely, hole signals display lower amplitudes, longer durations, and a peak towards the end of the transient. This occurs because holes possess a larger effective mass and drift in the opposite direction, accelerating as they enter regions of increasingly stronger electric field near the readout electrode, which maximizes the induced current just before collection.

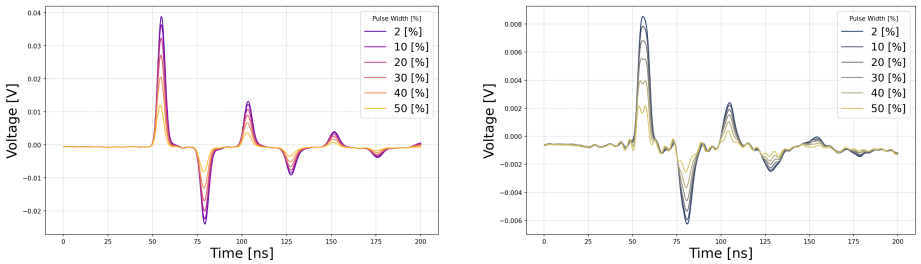
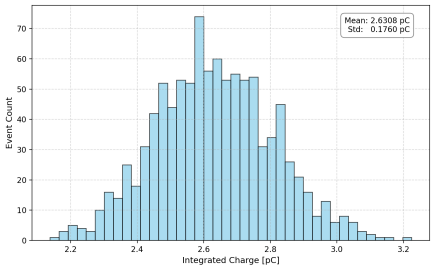


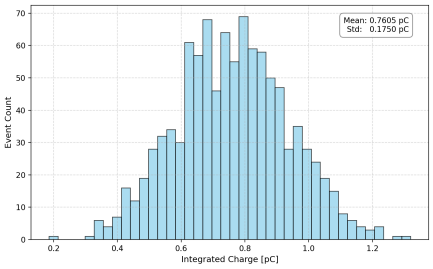
Figure 5: The mean waveform for different laser intensities with two types of charge carrier.

Furthermore, measurements were conducted at varying laser intensities while maintaining a constant reverse bias voltage of -500 V. As shown in Figure 5, the legend denotes the laser pulse width setting; in this specific configuration, increasing the pulse width percentage effectively reduces the laser intensity. The plots clearly demonstrate that as the laser intensity decreases, the amplitude of the induced signal proportionally decreases. This behavior is entirely expected from a physical standpoint. A lower laser intensity means that fewer photons are deposited onto the sensor surface per pulse. Consequently, fewer electron-hole pairs are generated in the depletion zone. Since the induced current is directly proportional to the total amount of moving charge (as dictated by the Shockley-Ramo theorem), a smaller generated charge naturally results in a lower voltage amplitude on the oscilloscope. It's for this same reason that also the signal durations shortens slightly. But this change is almost unnoticeable because the transit time depends mostly on the applied electric field and the carrier mobility, which were kept constant during this measurement.

In addition to analyzing the waveform shapes, the total gated charge of the signals was measured from the data acquisition for each waveform with different bias voltages. The measured charge was obtained by integrating the first peak of each waveform measurement, then those values were stored in histograms. Figure 6 presents examples of the resulting gated charge histograms for both electron and hole signals. The mean values and standard deviations extracted from these distributions provide a direct experimental measurement of the collected charge, which will be compared against the analytically computed charge in the following section.



(a) Charge histogram from electrons signals



(b) Charge histogram from holes signals

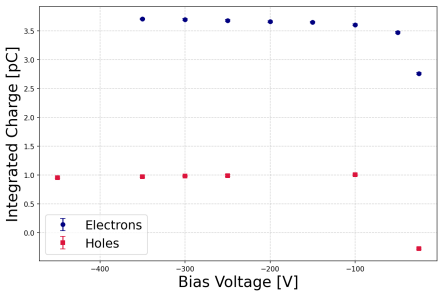
Figure 6: Histogram examples of measured charges from 1000 waveforms with two types of charge carrier.

3.2 Charge Comparison

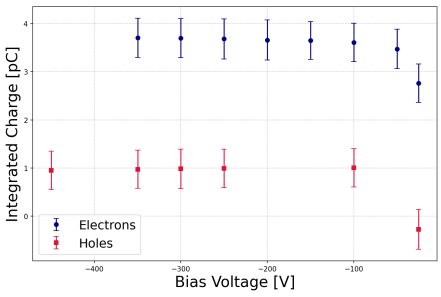
In this section, the analytically computed charge, derived from the averaged waveforms, is compared with the direct charge measurements obtained from the histograms from the previous section..

The computed charges were calculated by first converting the recorded voltage signals into current, assuming a standard resistance of 50 Ω for the whole measurement system ($I_i = V_i/50\,\Omega$). This current was then numerically integrated over the specific time interval $[t_s, t_e]$ corresponding to the primary signal peak:

$$Q = \sum_{i=t_s}^{t_e} I_i \Delta t_i \tag{10}$$



(a) Computed charges



(b) Measured charges

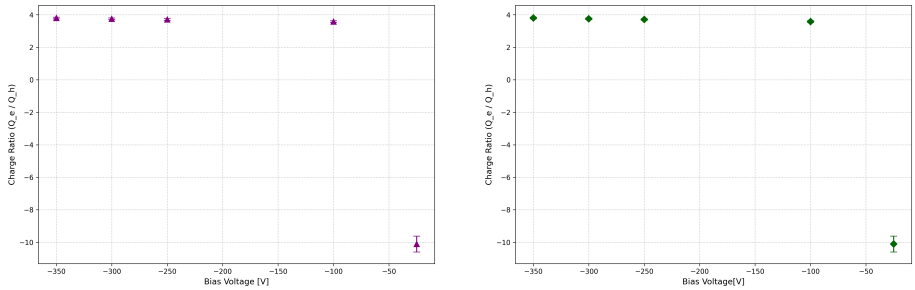
Figure 7: Comparison between computed and measured total collected charges as a function of the bias voltage.

Figure 7 displays the total collected charge for both electrons and holes as a function of the applied bias voltage. From these plots, it is evident that the

collected charge increases with voltage until it reaches a maximum plateau. This constant charge region corresponds to the sensor operating in the overdepletion regime, where full signal efficiency is achieved and the maximum generated charge is completely collected. At lower voltages, the collected charge appears smaller: this is primarily due to incomplete depletion which cause by dominant background electronic effects over the actual signals. This might be the reason of a negative charge value at -25V. Furthermore, the total charge magnitude for electrons is significantly higher than that for holes, although theoretically they should be the same. This discrepancy comes from the different illuminated sensor areas on the two sides of the detector, which are constrained by the physical sizes of the optical apertures in the setup. Overall, the analytically computed charges show excellent consistency with the direct measurements.

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To further investigate this area discrepancy, the ratio of the collected charges ($R_Q = Q_e/Q_h$) was evaluated. The charge ratio at different bias voltages is shown in Figure 8.

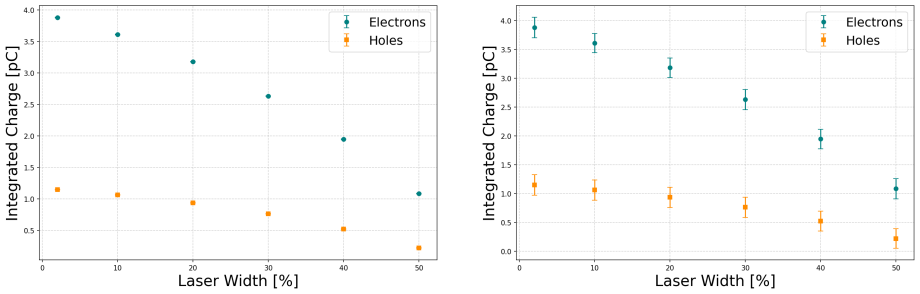


(a) Computed charge ratios (b) Measured charge ratios

Figure 8: Ratio of electron to hole charge as a function of bias voltage, comparing computed and measured data.

Based on the geometric difference between the optical apertures, the theoretically expected ratio is $R_A = 2.88 \pm 0.29$, from [1]. However, the plots reveal that at high bias voltages, the measured charge ratio stabilizes at a mean value of approximately $R_m = 3.71 \pm 0.08$. This experimental value is systematically higher and not fully consistent with the purely geometric theoretical prediction, suggesting that additional factors might have influenced the effective irradiated area.

Finally, Figure 9 illustrates the relationship between the collected charge and the laser intensity. A clear relation is observed for both electrons and holes. This behavior perfectly aligns with theoretical expectations: a higher laser intensity (given by lower pulse width, as explained before) corresponds to a proportionally larger amount of deposited energy, which in turn creates a greater number of electron-hole pairs in the bulk. As noted previously, the absolute charge values for electron signals remain systematically higher than those for holes solely due to the larger illuminated area on that specific side of the sensor.



(a) Computed charges (b) Measured charges

Figure 9: Comparison between computed and measured charge values as a function of laser intensity.

3.3 Charge Carrier Mobility Estimation

The final step of the data analysis involves estimating the effective mobility of the charge carriers. To achieve this, the signal duration (or transit time) for both electrons and holes was extracted from the recorded mean waveforms. The transit time was estimated by computing the Full Width at Half Maximum (FWHM) of the primary signal pulse for each applied bias voltage.

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As illustrated by the data points in Figure 10, there is an outlier at -25V due to waveform fluctuation and the peak was not clear resulting to difficulity to detemined FWHM.[2] The collection times systematically decrease as the applied bias voltage increases. This behavior is expected: a higher reverse bias voltage strengthens the internal electric field across the sensor. This stronger field accelerates the charge carriers, yielding higher drift velocities and consequently reducing the overall time required to cross the 300 μm silicon bulk. Furthermore, the plot highlights a clear distinction between the two carrier types: holes exhibit consistently longer transit times compared to electrons. This difference is a direct consequence of holes having a larger effective mass than electrons, which translates to a lower drift mobility within the crystal lattice.

To quantitatively estimate the mobility values, the experimental FWHM data were fitted as a function of the absolute bias voltage using the theoretical transit time models derived in Section 1:

$$T^{-} = \frac{d^2}{2\mu_e V_{dep}} \ln\left(\frac{V + V_{dep}}{V - V_{dep}}\right) \tag{11}$$

$$T^{+} = \frac{d^2}{2\mu_h V_{dep}} \ln\left(\frac{V + V_{dep}}{V - V_{dep}}\right) \tag{12}$$

where d is the sensor thickness, V_{dep} is the full depletion voltage, and μ_e and μ_h are the mobilities for electrons and holes, respectively.

[2] Note that the measurements done at lower bias voltages (e.g. -25 V) were excluded, since the sensor wasn't even in the depletion regime and they could alter the final fitting procedure.

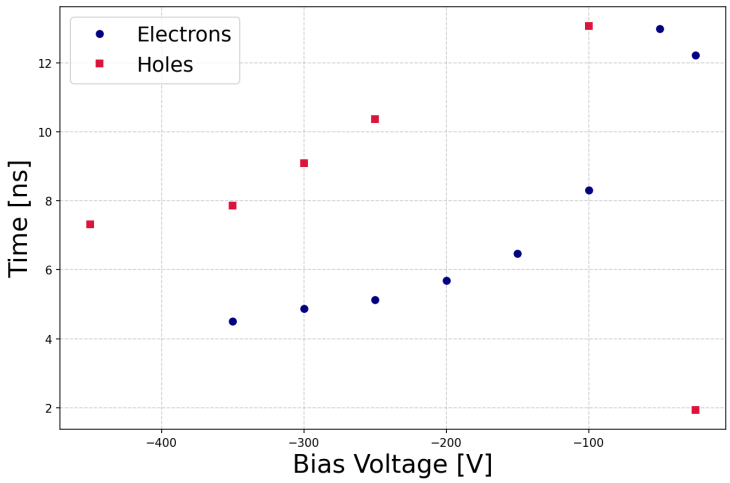


Figure 10: Transit time (FWHM) data points recorded as a function of the bias voltage for both electrons and holes.

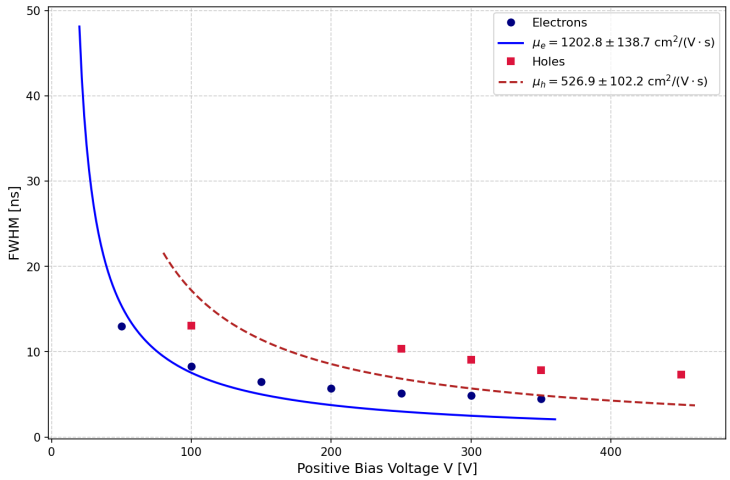


Figure 11: Analytical fit applied to the transit time (FWHM) as a function of the absolute positive bias voltage for both electrons and holes.

The fitting procedure, shown in Figure 11, yielded the following estimations for the effective mobilities:

- $\mu_e^{fit} = (1203 \pm 139)\,\mathrm{cm}^2/(\mathrm{V}\cdot\mathrm{s})$
- $\mu_h^{fit} = (527 \pm 102)\,\mathrm{cm}^2/(\mathrm{V}\cdot\mathrm{s})$

For standard high-purity silicon at room temperature ($T = 300$ K), the accepted theoretical mobility values are approximately:

- $\mu_e^{th} = 1400 - 1500\,\mathrm{cm}^2/(\mathrm{V}\cdot\mathrm{s})$
- $\mu_h^{th} = 450 - 500\,\mathrm{cm}^2/(\mathrm{V}\cdot\mathrm{s})$

The experimentally estimated hole mobility is well consistent with the theoretical expectations. The electron mobility, however, is slightly lower than the ideal value. Additionally, both of our estimations carry a relatively large statistical uncertainty. These discrepancies can be attributed to several experimental limitations. First, the limited number of acquired data points across the voltage range leads to a less constrained and less precise fit. Second, carrier mobilities are highly temperature-dependent; any deviation from an exact 300 K environment would alter the expected values. Finally, the theoretical models assume a perfectly uniform idealized crystal, whereas local variations in the effective doping concentration of the real sensor could introduce systematic deviations in the internal electric field, slightly altering the measured transit times.

4 Conclusions

The drift dynamics of charge carriers inside a silicon semiconductor sensor were investigated using the Transient-Current Technique. The transient signals generated by the drifting charges were collected and digitized under various experimental configurations, varying both the reverse bias voltage and the laser intensity. The observed waveform behaviors were highly consistent with theoretical models, allowing for the clear identification and separation of electron- and hole-dominated signals based on their distinct characteristics, such as amplitude, duration, and peak position.

Furthermore, the total collected charge was evaluated by comparing direct measurements (gated histograms) with the analytical integration of the averaged waveforms. The observed magnitude difference between the total electron and hole charges was explained by the different illuminated areas on the front and back sides of the sensor. Although the experimental charge ratio did not perfectly match the purely geometric theoretical prediction, likely due to secondary systematic effects, the deposited charge exhibited the expected linear dependence on the applied laser intensity.

Finally, the mobilities of both charge carriers were estimated by fitting the signal durations (FWHM) to theoretical drift models. The extracted values were $\mu_e = (1203 \pm 139)\,\mathrm{cm}^2/(\mathrm{V}\cdot\mathrm{s})$ for electrons and $\mu_h = (527 \pm 102)\,\mathrm{cm}^2/(\mathrm{V}\cdot\mathrm{s})$

for holes. While the hole mobility aligns well with theoretical predictions for high-purity silicon, the electron mobility is slightly lower than the ideal value. The relatively high statistical uncertainties and this slight deviation can be attributed to the limited number of data points used in the fitting procedure, potential temperature fluctuations inside the sensor during operation, and variations in the doping concentration that have modified the internal electric field. Overall, despite these experimental limitations, the results successfully validate the theoretical models of charge transport in semiconductor detectors.

References

[1] Advanced Laboratory Course in Particle Physics. *Measurement of the Drift of Charge Carriers in a Semiconductor Using the Transient-Current Technique: Experimental Instructions*. TU Dortmund, 2026.

[2] H. Kolanoski and N. Wermes. *Particle Detectors - Fundamentals and Applications*. Oxford University Press, 2022.